

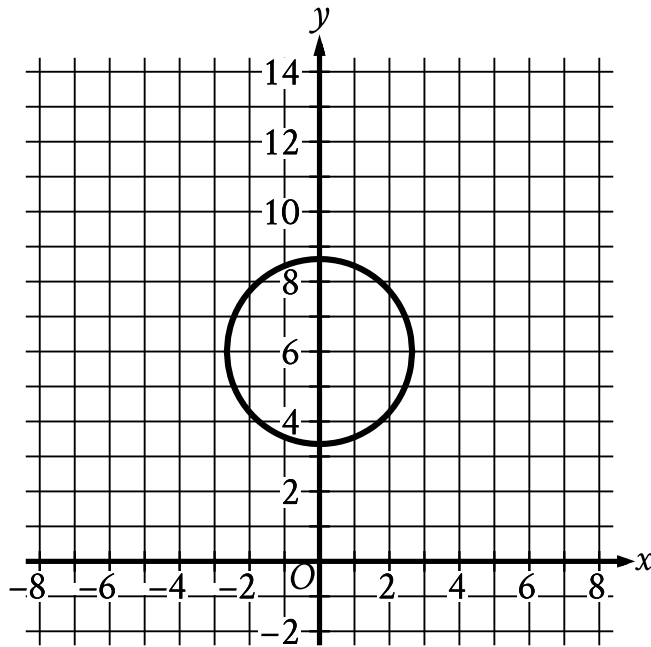
● HARD

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

08

10 questions · ~15 min



- The circle passes through the following approximate points:
 - (0 comma 8.6)
 - (2.6 comma 6)
 - (0 comma 3.4)
 - (negative 2.6 comma 6)

Circle A shown is defined by the equation $x^2 + y - 6^2 = 7$. Circle B (not shown) has the same radius but is translated 96 units to the right. If the equation of circle B is $x - h^2 + y - k^2 = a$, where h , k , and a are constants, what is the value of $4a$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

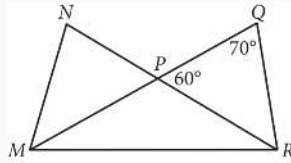
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

GEOMETRY AND TRIGONOMETRY

In triangle XYZ , angle Z is a right angle and the length of $\overline{YZ}$ is 24 units. If $\tan X = \frac{12}{35}$, what is the perimeter, in units, of triangle XYZ ?

- A. 188
- B. 168
- C. 84
- D. 71



In the figure above, $\overline{MQ}$ and $\overline{NR}$ intersect at point P, $NP = QP$, and $MP = PR$. What is the measure, in degrees, of $\angle QMR$? (Disregard the degree symbol when gridding your answer.)

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GEOMETRY AND TRIGONOMETRY

A right triangle has sides of length $2\sqrt{2}$, $6\sqrt{2}$, and $\sqrt{80}$ units. What is the area of the triangle, in square units?

- A. $8\sqrt{2} + \sqrt{80}$
- B. 12
- C. $24\sqrt{80}$
- D. 24

Q5

GEOMETRY AND TRIGONOMETRY

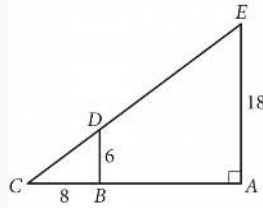
A circle in the xy-plane has equation $(x+3)^2 + (y-1)^2 = 25$. Which of the following points does NOT lie in the interior of the circle?

- A. $(-7, 3)$
- B. $(-3, 1)$
- C. $(0, 0)$
- D. $(3, 2)$

Q6

GRID-IN

GEOMETRY AND TRIGONOMETRY



In the figure above, $\overline{BD}$ is parallel to $\overline{AE}$. What is the length of $\overline{CE}$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

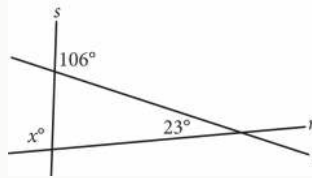
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

GRID-IN

GEOMETRY AND TRIGONOMETRY

Intersecting lines r , s , and t are shown below.



What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GEOMETRY AND TRIGONOMETRY

Two identical rectangular prisms each have a height of 90 centimeters cm. The base of each prism is a square, and the surface area of each prism is K cm². If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{92}{47}K$ cm². What is the side length, in cm, of each square base?

- A. 4
- B. 8
- C. 9
- D. 16

Q9

GEOMETRY AND TRIGONOMETRY

The equation $x^2 + y - 2^2 = 36$ represents circle A. Circle B is obtained by shifting circle A down 4 units in the xy -plane. Which of the following equations represents circle B?

- A. $x^2 + y + 2^2 = 36$
- B. $x^2 + y - 6^2 = 36$
- C. $x - 4^2 + y - 2^2 = 36$
- D. $x + 4^2 + y - 2^2 = 36$

Q10

GRID-IN

GEOMETRY AND TRIGONOMETRY

The perimeter of an equilateral triangle is 624 centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.